UNIVERSITY OF RAJSHAHI
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
B.Sc. Engineering Part III Even Semester Exam -2024
CSE 3222: Digital Signal Processing Lab.

SET A

Question 1: Demonstrate the sampling theorem, Nyquist rate, and aliasing using sinusoidal signals.

- (a) Consider $x_a(t) = 3\cos(100\pi t)$. Plot the sampled signal for $f_s = 400$ Hz, 100 Hz, and 60 Hz, and observe aliasing.
- (b) Consider $x_a(t) = 3\cos(2000\pi t) + 5\sin(6000\pi t) + 10\cos(12000\pi t)$. Determine the maximum frequency and Nyquist rate, then sample the signal at $f_s = 15000$ Hz.

Question 2: Compute the linear convolution of two discrete-time signals using both built-in and manual methods.

- (a) Given $x[n] = a^n u[n]$ and $h[n] = a^n u[n]$, where $a = 0.5$, calculate $y[n] = x[n] * h[n]$ for $n = 0$ to 10. Display the input, impulse response, output table, and stem plots.
- (b) Implement manual convolution using nested loops without using np.convolve.
- (c) Calculate the convolution of $x = [1,2,3]$ and $h = [1,1,1]$ using np.convolve.
- (d) Calculate the same convolution manually and display the step-by-step calculations for each output sample.

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